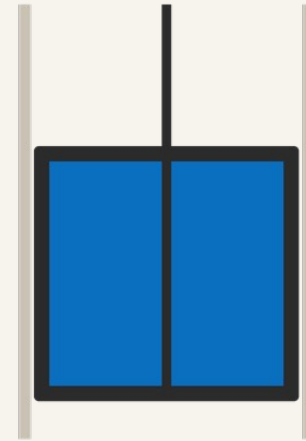


OUR DYNAMIC UNIVERSE 2 · FORCES, ENERGY & POWER

Forces, Energy & Power

What makes things accelerate, and what it costs to make it happen.

Higher Physics



Starter

2012 Revised Higher Physics

Scan the code and answer these:

Q17

Q18

Q20

Section A · e.m.f., current, p-n
junctions
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 1

Balanced and unbalanced forces

An object's motion only ever changes because the forces on it stop balancing.

By the end of this part...

- 1 I can add forces as vectors to find the resultant force
- 2 I can state Newton's first law and use it to explain motion
- 3 I can use $F = ma$ to solve problems involving unbalanced forces

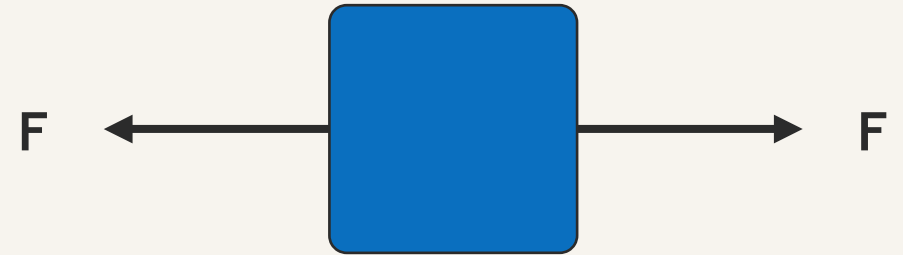
Balanced and unbalanced forces

- Forces are vectors — combine them by vector addition, not by simply adding sizes.
- Balanced forces have a resultant of zero — no change in motion.
- Unbalanced forces have a resultant that is not zero — the object accelerates.

Newton's first law

An object stays at rest, or moving at constant velocity in a straight line, unless acted on by an unbalanced force.

The two forces shown here are equal and opposite — balanced — so this object's velocity is not changing.



$$\mathbf{F = ma}$$

$$F = ma$$

F

unbalanced force
(N)

m

mass (kg)

a

acceleration (m
 s^{-2})

F is the RESULTANT (unbalanced) force — if several forces act, add them as vectors first.

Worked example – unbalanced force

A trolley of mass 2.0 kg is pulled by a force of 6.0 N against 2.0 N of friction. Find its acceleration.

$$F = 6.0 \text{ N}$$

$$f = 2.0 \text{ N}$$

$$m = 2.0 \text{ kg}$$

$$a = ?$$

$$F = 6.0 - 2.0$$

$$F = 4.0 \text{ N}$$

$$F = ma$$

$$4.0 = 2.0 \times a$$

$$\underline{a = 2.0 \text{ m s}^{-2}}$$



Forces and Newton's first law

Forces are vectors – combine them by vector addition.

Balanced forces (resultant = 0): no change in motion.

Newton's first law: an object stays at rest, or at constant velocity, unless acted on by an unbalanced force.

$$F = ma$$

Your turn – balanced forces

SHOW ME BOARDS

- 1 A book on a table has weight 5 N. State the size of the normal force from the table, and explain your reasoning.
- 2 A box of mass 4.0 kg accelerates at 1.5 m s^{-2} . Find the unbalanced force acting on it.
- 3 State Newton's first law in your own words.

TRY THIS – Answers

- 1 5 N — the forces are balanced, so the book is not accelerating.
- 2 6.0 N.
- 3 An object stays at rest or at constant velocity unless an unbalanced force acts on it.

Starter

2015 Higher Physics

Scan the code and answer these:

Q8

Q10

Q16

Section A · particles, a.c., uncertainty
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 2

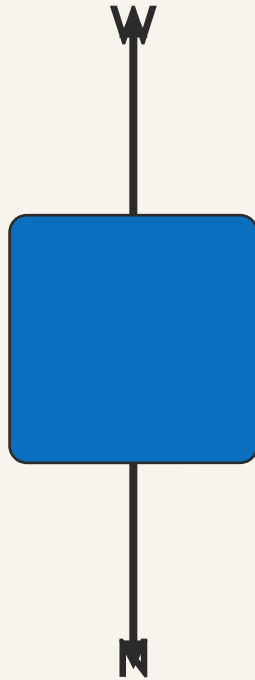
Newton's laws in action

Systems of connected objects still obey $F = ma$ — you just have to be careful what "m" means.

By the end of this part...

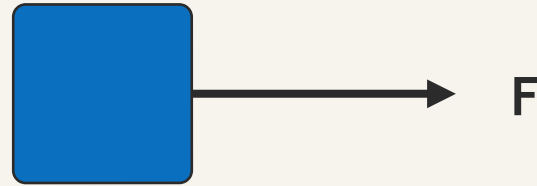
- 1 I can use Newton's second law to explain the motion of an object
- 2 I can draw a free body diagram for a simple system
- 3 I can solve problems involving connected objects and tension

Free body diagrams



A free body diagram shows **ONLY** the forces acting **ON** one object — never the forces it exerts on anything else. Draw every force as an arrow from the object's centre, sized roughly by strength and labelled by name. Here: weight W (down) and the normal force N (up) — for an object at rest on a surface, these are equal and balanced.

Connected objects



Two objects joined by a string act as ONE system for the whole force, but each object still obeys $F = ma$ on its own – the string's tension is the force between them.

Worked example – connected objects

Two blocks of mass 4.0 kg and 6.0 kg, joined by a string on a frictionless surface, are pulled by a force of 20 N. Find the acceleration of the system and the tension in the string.

$$F = 20 \text{ N}$$

$$m_1 = 4.0 \text{ kg}$$

$$m_2 = 6.0 \text{ kg}$$

$$a, T = ?$$

$$F = (m_1 + m_2)a$$

$$20 = 10.0 \times a$$

$$T = m_1 a$$

$$\underline{T = 4.0 \times 2.0 = 8.0 \text{ N}}$$

$$a = 2.0 \text{ m s}^{-2}$$

Watch Out – tension problems

THE MISTAKE

- Forgetting friction when finding the unbalanced force
- Using the total mass to find the tension on one block

THE CORRECT ANSWER

- Add up ALL the forces acting along the line of motion first.
- Use ONE block's own $F = ma$, not the whole system's.

Your turn – Newton's laws

SHOW ME BOARDS

- 1 State what a free body diagram shows, and what it must never include.
- 2 Two trolleys (2.0 kg, 3.0 kg) joined by a string are pulled by 10 N on a frictionless bench. Find the acceleration.
- 3 For the trolleys above, find the tension in the string.

TRY THIS – Answers

- 1 Only the forces acting ON that one object – never the forces it exerts on something else.
- 2 2.0 m s^{-2} .
- 3 6.0 N (from the 3.0 kg block: $T = 3.0 \times 2.0$).

Starter

2010 Higher Physics

Scan the code and answer these:

Q2

Q16

Section A · motion, semiconductors
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 3

Friction and terminal velocity

Friction always opposes motion — and for a falling object, it can grow until the forces balance again.

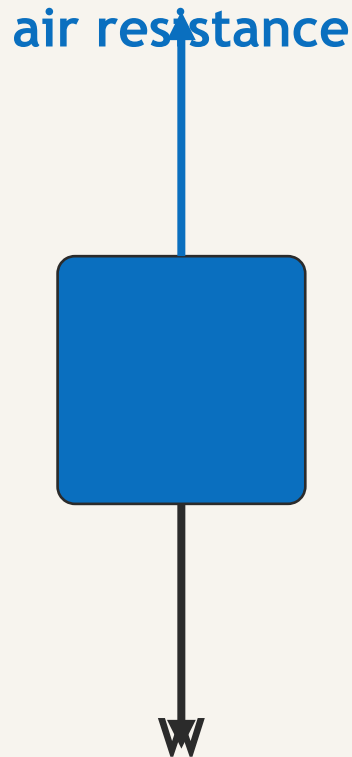
By the end of this part...

- 1 I can describe the effect of friction on a moving object
- 2 I can explain, in terms of forces, why a falling object reaches terminal velocity

Friction

- Friction is a force that opposes the relative motion between two surfaces in contact.
- It always acts in the direction that would slow the moving object down.
- Friction depends on the surfaces and how hard they press together — no static/dynamic distinction is needed at this level.

Terminal velocity



A falling object speeds up, so air resistance (which grows with speed) increases too.

Eventually air resistance equals the object's weight — the forces balance.

The object then falls at a constant maximum speed — its terminal velocity.

Watch Out – terminal velocity

THE MISTAKE

- The object stops accelerating, not moving
- Acceleration increases as terminal velocity nears

THE CORRECT ANSWER

- Balanced forces mean constant speed, not zero.
- Air resistance grows to match the weight, so a falls to zero.

Your turn – friction and terminal velocity

SHOW ME BOARDS

- 1 State the direction friction acts in, relative to an object's motion.
- 2 A skydiver's weight is 800 N. At terminal velocity, what is the air resistance acting on them?
- 3 Explain why a skydiver's acceleration decreases as they fall.

TRY THIS – Answers

- 1 Opposite to the direction of motion.
- 2 800 N – the forces are balanced at terminal velocity.
- 3 As speed increases, air resistance increases, so the unbalanced force – and therefore the acceleration – falls.

Starter

2013 Revised Higher Physics

Scan the code and answer these:

Q4

Q5

Q20

Section A · connected objects, resolving
Show your working, not just the letter.



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Marking scheme**

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PART 4

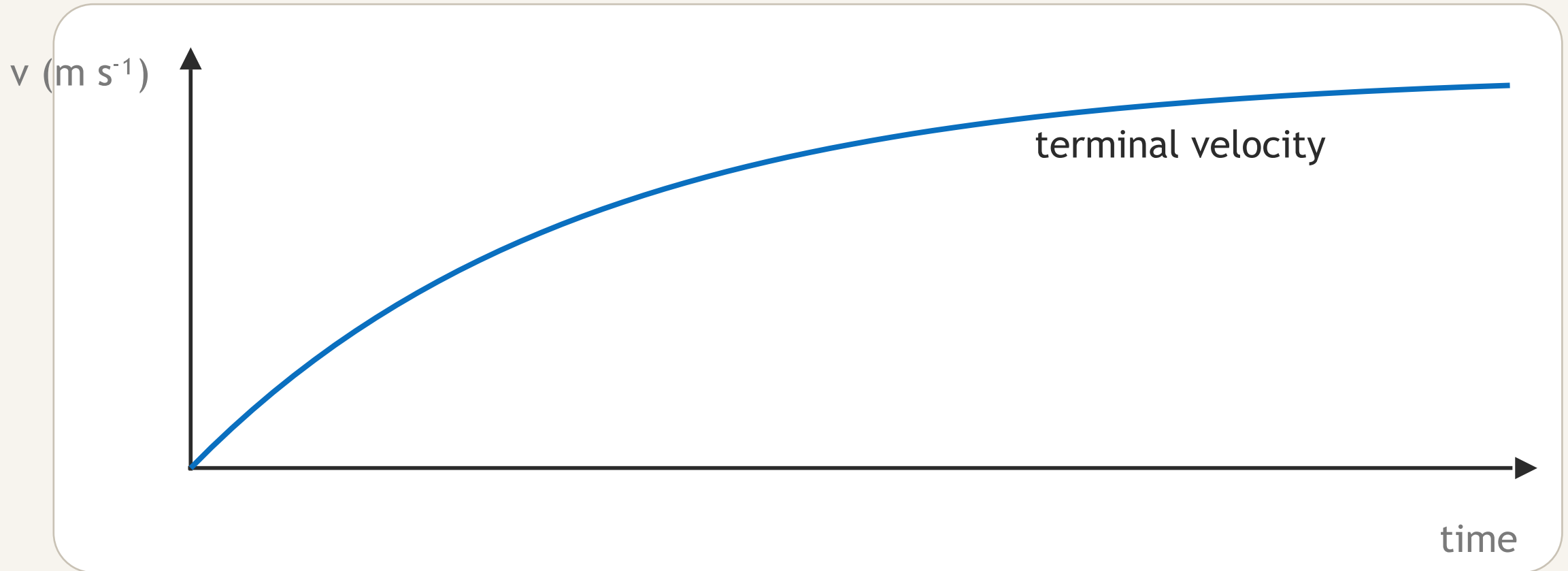
Falling with air resistance

A real velocity-time graph for a falling object tells the whole story of the forces acting on it.

By the end of this part...

- 1 I can interpret a velocity-time graph for a falling object with air resistance
- 2 I can relate the shape of the graph to the changing forces

A skydiver's velocity-time graph



Reading the graph

- Steep at the start — weight far exceeds air resistance, so acceleration is close to g .
- The gradient falls as speed builds — air resistance is growing, so the unbalanced force is shrinking.
- The graph goes flat once air resistance equals weight — terminal velocity, zero acceleration.

Worked example – a real skydiver's data

A skydiver's speed is recorded every 3 s after jumping: 0, 25, 45, 50, 53, 55, 55, 55 m s⁻¹ (0 to 21 s). State what is true about the forces between 15 s and 21 s, and about the motion 0-9 s.

$$v(15\text{s}) = 55 \text{ m s}^{-1}$$

$$v(18\text{s}) = 55 \text{ m s}^{-1}$$

$$v(21\text{s}) = 55 \text{ m s}^{-1}$$

constant speed 15-21 s

⇒ forces are BALANCED in this interval

0-9 s: speed rises but by less each 3 s

acceleration is falling, not rising

Watch Out — reading v-t graphs with air resistance

THE MISTAKE

- Right graph shape, wrong acceleration-time pairing
- Assuming vertical acceleration keeps rising

THE CORRECT ANSWER

- Check the gradient at the SAME points on both graphs.
- Acceleration falls to zero near terminal velocity.

Your turn – falling with air resistance

SHOW ME BOARDS

- 1 Sketch the velocity-time graph for an object falling from rest and reaching terminal velocity.
- 2 State what happens to the gradient of the graph as the object approaches terminal velocity, and why.
- 3 State the relationship between weight and air resistance once terminal velocity is reached.

TRY THIS – Answers

- 1 A curve, steep at first, levelling off to a flat (horizontal) line.
- 2 The gradient falls towards zero, because air resistance grows and the unbalanced force shrinks.
- 3 They are equal.

Starter

2014 Revised Higher Physics

Scan the code and answer these:

Q2

Q17

Q20

Section A · motion, p.d./power

Show your working, not just the letter.



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Marking scheme**

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PART 5

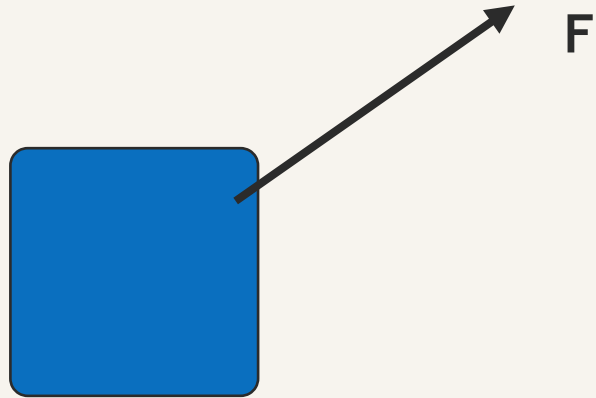
Resolving vectors

Any vector can be split into two components at right angles to each other — the reverse of adding forces up.

By the end of this part...

- 1 I can resolve a vector into two perpendicular components

Resolving a force



$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

θ is measured from the horizontal — the SAME resolving idea used for projectiles in ODU 1, just applied to a force instead of a velocity.

Worked example – resolving a force

A force of 15 N acts on a box at 60° above the horizontal.
Find the horizontal and vertical components of the force.

$$F = 15 \text{ N}$$

$$\theta = 60^\circ$$

$$F_x, F_y = ?$$

$$F_x = F \cos \theta$$

$$F_x = 15 \times \cos 60^\circ$$

$$F_x = 7.5 \text{ N}$$

$$F_y = F \sin \theta$$

$$F_y = 15 \times \sin 60^\circ$$

$$\underline{F_y = 13.0 \text{ N}}$$



Resolving a vector

Any vector at angle θ to the horizontal splits into two perpendicular components:

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

Your turn – resolving vectors

SHOW ME BOARDS

- 1 A force of 20 N acts at 30° above the horizontal. Find its horizontal component.
- 2 For the same force, find its vertical component.
- 3 State which trig function pairs with which component when the angle is measured from the horizontal.

TRY THIS – Answers

- 1 17.3 N.
- 2 10.0 N.
- 3 cos with the horizontal component, sin with the vertical – reversed if the angle is measured from the vertical instead.

Starter

2009 Higher Physics

Scan the code and answer these:

Q2

Q9

Q18

Section A · motion, semiconductors
Show your working, not just the letter.



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Marking scheme**

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PART 6

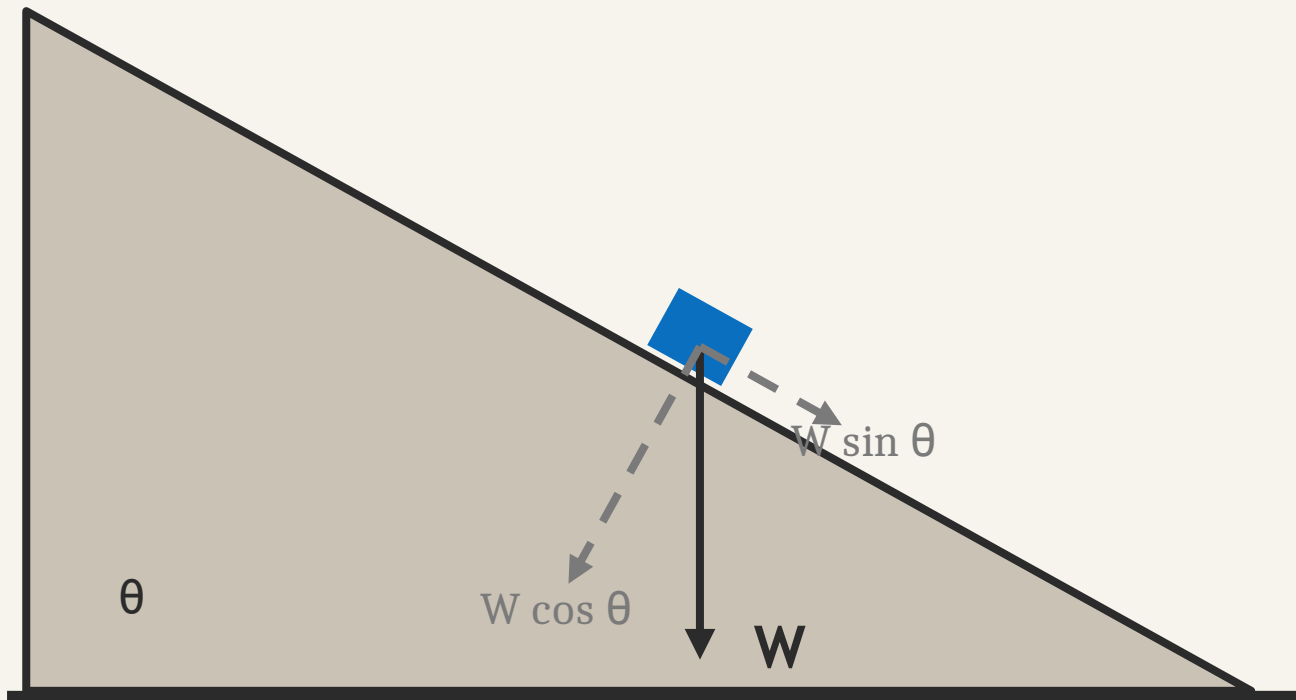
Forces on a slope

On an incline, weight itself has to be resolved before $F = ma$ can be used at all.

By the end of this part...

- 1 I can resolve the weight of an object on a slope into components parallel and normal to the slope
- 2 I can solve problems involving forces, including friction, on a slope

Weight on a slope



$W \sin \theta$ — parallel to the slope.

$W \cos \theta$ — normal to the slope.

$W \sin \theta$ pulls the object **DOWN** the slope; $W \cos \theta$ presses it **INTO** the slope, balanced by the normal force.

Worked example – box pushed up a slope

A 5.0 kg box is pushed up a frictionless 30° slope by a force of 40 N parallel to the slope. Find its acceleration up the slope.

$$m = 5.0 \text{ kg}$$

$$\theta = 30^\circ$$

$$F = 40 \text{ N (up-slope)}$$

$$W = mg = 5.0 \times 9.8 = 49 \text{ N}$$

$$W \sin \theta = 49 \times \sin 30^\circ = 24.5 \text{ N}$$

$$F = ma$$

$$40 - 24.5 = 5.0 \times a$$

$$\underline{a = 3.1 \text{ m s}^{-2}}$$

Watch Out — friction on a slope

THE MISTAKE

- Subtracting friction when moving DOWN the slope
- Not resolving the weight before finding F

THE CORRECT ANSWER

- Moving down: friction acts up-slope, so SUBTRACT it.
- Resolve the weight first — $F = ma$ needs $W \sin \theta$, not W .

Your turn – forces on a slope

SHOW ME BOARDS

- 1 A 2.0 kg box rests on a frictionless slope at 20° . Find the component of its weight acting down the slope.
- 2 State which component of weight is balanced by the normal force on a slope.
- 3 A box slides DOWN a slope. State the direction friction acts in.

TRY THIS – Answers

- 1 6.7 N.
- 2 The component normal to the slope, $W \cos \theta$.
- 3 Up the slope – opposing the motion.

Starter

Revised Higher Specimen

Scan the code and answer these:

Q1

Q7

Q14

Section A · motion, forces/energy
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 7

Work done

Every energy calculation in this course starts from the same idea: force applied over a distance.

By the end of this part...

- 1 I can use $E_w = Fd$ to solve problems involving work done

Work done

$$E_w = Fd$$

E_w

work done (J)

F

force (N)

d

distance (m)

Work is done only while the force and the movement are in the SAME direction — use the component of F along the motion.

Worked example – work done

A force of 25 N pushes a crate 4.0 m across a warehouse floor.
Find the work done.

$$F = 25 \text{ N}$$

$$d = 4.0 \text{ m}$$

$$E_w = ?$$

$$E_w = Fd$$

$$E_w = 25 \times 4.0$$

$$\underline{E_w = 100 \text{ J}}$$

Your turn — work done

SHOW ME BOARDS

- 1 A force of 12 N moves a box 3.0 m. Find the work done.
- 2 State the unit work done is measured in.
- 3 A force acts at an angle to an object's motion. Explain what changes in the $E_w = Fd$ calculation.

TRY THIS – Answers

- 1 36 J.
- 2 The joule (J).
- 3 Only the component of the force **ALONG** the direction of motion does work – resolve the force first.

Starter

2007 Higher Physics

Scan the code and answer these:

Q2

Q9

Q10

Section A · motion, p.d./power, a.c.
Show your working, not just the letter.



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Marking scheme**

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PART 8

Potential and kinetic energy

Height stores energy; speed carries it — two ways of asking the same question.

By the end of this part...

- 1 I can use $E_p = mgh$ to solve problems involving potential energy
- 2 I can use $E_k = \frac{1}{2}mv^2$ to solve problems involving kinetic energy

Potential energy

$$E_p = mgh$$

E_p

gravitational P.E.
(J)

m

mass (kg)

h

height (m)

h is measured from whatever level you choose as zero — only the CHANGE in height usually matters.

Worked example – potential energy

A ball of mass 0.50 kg is raised 2.0 m. Find its gain in P.E.

$$m = 0.50 \text{ kg}$$

$$h = 2.0 \text{ m}$$

$$E_p = ?$$

$$E_p = mgh$$

$$E_p = 0.50 \times 9.8 \times 2.0$$

$$\underline{E_p = 9.8 \text{ J}}$$

Kinetic energy

$$E_k = \frac{1}{2} m v^2$$

E_k

kinetic energy (J)

m

mass (kg)

v

speed (m s⁻¹)

Kinetic energy depends on v^2 , so doubling the speed **QUADRUPLES** the kinetic energy.

Worked example – kinetic energy

A car of mass 1000 kg travels at 20 m s^{-1} . Find its kinetic energy.

$$m = 1000 \text{ kg}$$

$$v = 20 \text{ m s}^{-1}$$

$$E_k = ?$$

$$E_k = \frac{1}{2} m v^2$$

$$E_k = 0.5 \times 1000 \times 20^2$$

$$\underline{E_k = 200\,000 \text{ J}}$$



Potential and kinetic energy

$$E_p = mgh$$

$$E_k = \frac{1}{2} m v^2$$

Doubling v quadruples E_k , since E_k depends on v^2 .

Your turn – potential and kinetic energy

SHOW ME BOARDS

- 1 A 2.0 kg mass is raised 3.0 m. Find its gain in potential energy.
- 2 A 0.20 kg ball moves at 8.0 m s^{-1} . Find its kinetic energy.
- 3 A cyclist doubles their speed. State the factor their kinetic energy increases by.

TRY THIS – Answers

- 1 58.8 J.
- 2 6.4 J.
- 3 4 times.

Starter

2008 Higher Physics

Scan the code and answer these:

Q2

Q3

Section A · projectiles, forces/energy
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 9

Conservation of energy

Energy is never created or destroyed — potential energy lost becomes kinetic energy gained.

By the end of this part...

- 1 I can use the principle of conservation of energy to solve problems involving E_p and E_k

Conservation of energy

- Energy cannot be created or destroyed — only transferred from one form to another.
- For a falling (or swinging) object with no friction or air resistance:
 $\text{P.E. lost} = \text{K.E. gained}$.
- This lets you find a speed from a height, or a height from a speed, without knowing the time taken.

Worked example – conservation of energy

A ball of mass 0.20 kg falls from rest through 5.0 m. Find its speed just before hitting the ground (ignore air resistance).

$$m = 0.20 \text{ kg}$$

$$h = 5.0 \text{ m}$$

$$v = ?$$

$$E_p \text{ lost} = E_k \text{ gained}$$

$$mgh = \frac{1}{2} m v^2$$

$$gh = \frac{1}{2} v^2 \quad (\text{m cancels})$$

$$9.8 \times 5.0 = 0.5 \times v^2$$

$$\underline{v = 9.9 \text{ m s}^{-1}}$$

Watch Out – conservation of energy

THE MISTAKE

- Doubling the release height doubles the final speed
- Forgetting mass cancels out of a P.E.-to-K.E. conversion

THE CORRECT ANSWER

- Doubling h needs $\sqrt{2}$ for speed – $E_p \propto h$ but $E_k \propto v^2$.
- $mgh = \frac{1}{2}mv^2$ – mass cancels out of the final speed.

Your turn – conservation of energy

SHOW ME BOARDS

- 1 A ball falls from rest through 2.0 m. Find its speed just before landing (ignore air resistance).
- 2 State the energy transformation taking place as the ball falls.
- 3 A second, heavier ball falls from the same height. State how its final speed compares to the first ball's.

TRY THIS – Answers

- 1 6.3 m s^{-1} .
- 2 Gravitational potential energy transforms to kinetic energy.
- 3 The same – mass cancels, so final speed does not depend on it.

Starter

2011 Higher Physics

Scan the code and answer these:

Q2

Q12

Q13

Section A · graphs, a.c., uncertainty
Show your working, not just the letter.



Scan for the paper

No marking scheme

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PART 10

Power

Power is how quickly work gets done — the same energy, delivered faster, is more power.

By the end of this part...

- 1 I can use $P = E_w / t$ to solve problems involving power

Power

$$P = E_w/t$$

P

power (W)

E_w

energy/work (J)

t

time (s)

1 watt = 1 joule of energy transferred per second.

Worked example – power

A motor does 5000 J of work in 20 s. Find its power.

$$E_w = 5000 \text{ J}$$

$$t = 20 \text{ s}$$

$$P = ?$$

$$P = E_w / t$$

$$P = 5000 / 20$$

$$\underline{P = 250 \text{ W}}$$



Work, energy and power – summary

$$E_w = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2} m v^2$$

$$P = E_w/t$$

Checkpoint

- 1 State Newton's first law, and use it to explain balanced forces.
- 2 A box is pushed up a frictionless slope. Sketch the forces acting on it, including the resolved components of its weight.
- 3 State the relationship between weight and air resistance at terminal velocity.
- 4 State the principle of conservation of energy, and use it in a P.E.-to-K.E. problem.

Exam Practice

2025 Q4 — forces and friction on a slope.
2024 Q4 — a skydiver's velocity-time graph.
2023 Q4 — conservation of energy, a
pendulum.

Past papers, Section A

LOOKING AHEAD

Next: Momentum & Collisions

What happens when moving objects meet — and why a force-time graph is the key to it.

Plenary

- 1 State Newton's first and second laws.
- 2 Sketch a free body diagram for an object at terminal velocity.
- 3 Resolve a 10 N force at 45° above the horizontal into components.
- 4 State the four energy/power relationships from this deck.
- 5 Explain, using conservation of energy, why a pendulum's speed is greatest at the bottom of its swing.

- 1 Explain why the normal force on a slope is less than the object's full weight.
- 2 Two skiers, one heavier, start from rest at the top of the same slope. Explain why they reach the bottom at the same speed.

Extension

EXTRA PRACTICE

- 1 A car's engine delivers constant power. Explain, using $P = E_w/t$ and $F = ma$, why its acceleration falls as its speed increases.
- 2 Derive weight = air resistance at terminal velocity from Newton's first law.

Extension — EXTRA

EXTRA PRACTICE

1

Research the terminal velocity of a skydiver in different body positions, and explain the difference using this deck's ideas.